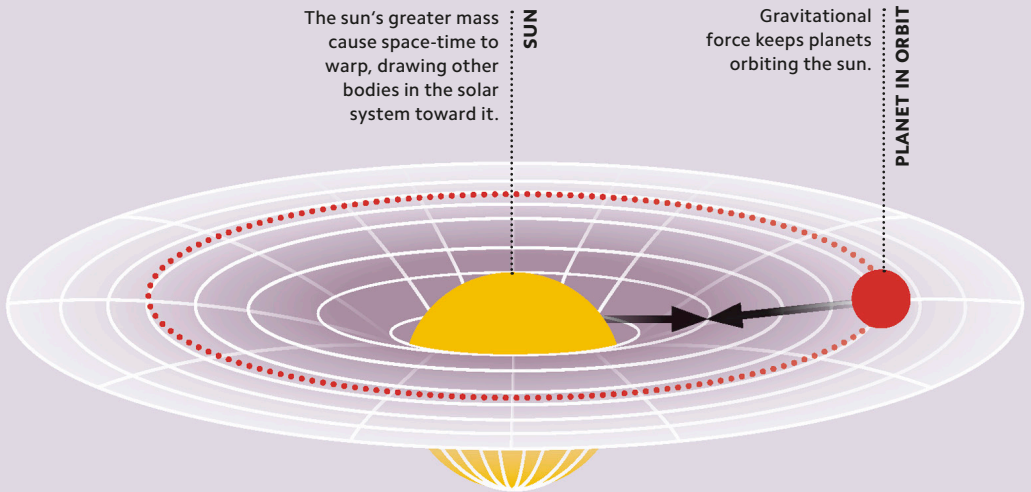


DRAWN TOGETHER

Gravitation is an attractive force between objects with mass. It is extremely weak and only becomes apparent between objects of large mass, yet, like electromagnetism, it has an infinite range. The best model for understanding gravitation is Einstein's General Relativity, a theory that seems completely separate from quantum physics. Understanding how gravity works on the level of particles poses many baffling questions (see pp.136–45).



The sun's greater mass cause space-time to warp, drawing other bodies in the solar system toward it.

SUN

Gravitational force keeps planets orbiting the sun.

PLANET IN ORBIT

Space-time

Einstein described the three dimensions of space and the dimension of time as a four-dimensional grid called space-time. General Relativity explains gravity as arising from distortions in space-time by massive objects.